14. Use induction to prove that $10^n + 3 \times 4^{n+2} + 5$, is divisible by 9, for all natural numbers n.
15. Prove that using the Mathematical induction

$$\sin(\alpha) + \sin\left(\alpha + \frac{\pi}{6}\right) + \sin\left(\alpha + \frac{2\pi}{6}\right) + \cdots + \sin\left(\alpha + \frac{(n-1)\pi}{6}\right) = \frac{\sin\left(\alpha + \frac{(n-1)\pi}{12}\right) \times \sin\left(\frac{n\pi}{12}\right)}{\sin\left(\frac{\pi}{12}\right)}.$$

LWEO3

Exercise - 4.5

Choose the correct or the most suitable answer

1. The sum of the digits at the 10th place of all numbers formed with the help of 2, 4, 5, 7 taken all at a time is

   (1) 432 (2) 108 (3) 36 (4) 18

2. In an examination there are three multiple choice questions and each question has 5 choices . Number of ways in which a student can fail to get all answer correct is

   (1) 125 (2) 124 (3) 64 (4) 63

3. The number of ways in which the following prize be given to a class of 30 boys first and second in mathematics, first and second in physics, first in chemistry and first in English is

   (1) $30^4 \times 29^2$ (2) $30^3 \times 29^3$ (3) $30^2 \times 29^4$ (4) 30×29^5.

4. The number of 5 digit numbers all digits of which are odd is

   (1) 25 (2) 5^5 (3) 5^6 (4) 625.

5. In 3 fingers, the number of ways four rings can be worn is _____ ways.

   (1) $4^3 - 1$ (2) 3^4 (3) 68 (4) 64

6. If ${}^{(n+5)}P_{(n+1)} = (\frac{11(n-1)}{2})^{(n+3)}P_n$, then the value of n are

   (1) 7 and 11 (2) 6 and 7 (3) 2 and 11 (4) 2 and 6.

7. The product of r consecutive positive integers is divisible by

   (1) $r!$ (2) $(r-1)!$ (3) $(r+1)!$ (4) r^r.

8. The number of five digit telephone numbers having at least one of their digits repeated is

   (1) 90000 (2) 10000 (3) 30240 (4) 69760.

9. If ${}^{a^2-a}C_2 = {}^{a^2-a}C_4$ then the value of 'a' is

   (1) 2 (2) 3 (3) 4 (4) 5

10. There are 10 points in a plane and 4 of them are collinear. The number of straight lines joining any two points is

    (1) 45 (2) 40 (3) 39 (4) 38.

11. The number of ways in which a host lady invite 8 people for a party of 8 out of 12 people of whom two do not want to attend the party together is

    (1) $2 \times {}^{11}C_7 + {}^{10}C_8$ (2) ${}^{11}C_7 + {}^{10}C_8$ (3) ${}^{12}C_8 - {}^{10}C_6$ (4) ${}^{10}C_6 + 2!$.